

UNIVERSITY OF AMSTERDAM

BSc ECONOMETRICS AND DATA SCIENCE THESIS

When to Optimize and When to Diversify: A Regime-Switching Approach to Portfolio Construction

Kiril Ivanov

Supervisor: Sanders Barendse

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Abstract

This is a short summary of your thesis. State the problem, methods, key results, and conclusions. (Max 300 words.)

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Chapter 1

Introduction

Background, problem statement, research questions, and thesis outline.

Chapter 2

Literature Review

This chapter builds up to the thesis in four steps. Sections 2.1 and 2.2 introduce Modern Portfolio Theory and the estimation problems that limit its practical value. Sections 2.3 and 2.4 review the two main responses to those problems: robust covariance estimation, which sharpens the inputs to mean-variance optimisation, and hierarchical methods, which sidestep optimisation entirely. Section 2.5 turns to the detection of market regimes through Hidden Markov Models, and Section 2.6 surveys the still-sparse literature on regime-conditional portfolio construction. Section 2.7 pulls these threads together and states the gap that this thesis addresses.

2.1 Classical portfolio theory

Modern Portfolio Theory begins with Markowitz [1952]. Markowitz translated the intuition that investors care about both expected return and risk into a tractable optimisation problem. With $\mu \in \mathbb{R}^N$ the vector of expected asset returns and Σ the $N \times N$ covariance matrix of those returns, the mean-variance optimal portfolio w^* solves

$$\min_w w^\top \Sigma w \quad \text{s.t.} \quad w^\top \mu \geq \mu^*, \quad \mathbf{1}^\top w = 1. \quad (2.1)$$

As the target return μ^* varies, the solutions trace out the efficient frontier, and each investor picks the point on it that matches their own risk preference.

Three classical results extended this framework. Tobin [1958] proved the two-fund separation theorem: once a risk-free asset exists, every investor holds the same risky portfolio, scaled to preference. Sharpe [1964] pushed Markowitz's logic into equilibrium, deriving the Capital Asset Pricing Model and linking expected returns to systematic risk. Merton [1972] wrote the efficient frontier analytically and described its geometry.

The appeal of MPT is its clarity, but the appeal rests on assumptions that are routinely violated: stationary return moments, Gaussian returns, and — the crux of this thesis — the assumption that μ and Σ are known rather than estimated. These gaps drive most of the subsequent literature.

2.2 The estimation problem

In practice μ and Σ are replaced by sample estimates, and the replacement turns out to be consequential. Michaud [1989] named the result the “Markowitz optimisation enigma”: the same procedure that delivers the in-sample optimal portfolio also maximises estimation error out of sample, producing extreme and unstable weights. Best and Grauer [1991] showed analytically that a small perturbation in the expected-return vector produces a large shift in w^* , and Chopra and Ziemba [1993] quantified the relative damage: errors in $\hat{\mu}$ drive out-of-sample losses roughly an order of magnitude more than errors in $\hat{\Sigma}$.

The most influential empirical verdict came from DeMiguel et al. [2009], who compared fourteen sophisticated portfolio construction methods against the naive $1/N$ rule on seven datasets. None of them consistently beat the equal-weight benchmark out of sample once estimation error was accounted for. Their implication — that at realistic sample sizes, “optimisation” can be an expensive way of fitting noise — remains the benchmark any new method must clear.

Together these findings define the *estimation problem*: any rule that amplifies input noise tends to underperform simpler alternatives, and mean-variance optimisation amplifies noise precisely because it relies on the inverse covariance matrix. The literature has since moved in two directions. One line tries to fix the inputs; the other bypasses optimisation altogether.

2.3 Robust covariance estimation

The first response attacks the covariance matrix directly. Ledoit and Wolf [2003, 2004] proposed the *linear shrinkage* estimator

$$\hat{\Sigma}_{\text{LW}} = \alpha^* F + (1 - \alpha^*) S, \quad (2.2)$$

which convexly combines the sample covariance S with a low-dimensional structured target F — most commonly the constant-correlation target. The shrinkage intensity $\alpha^* \in [0, 1]$ is picked in closed form by minimising the expected Frobenius distance to the true covariance. The resulting estimator is well-conditioned and invertible even in small samples, and it has become the default covariance estimator in portfolio applications. Ledoit and Wolf [2020] later introduced *nonlinear shrinkage*, which shrinks each eigenvalue of S individually under a rule derived from random-matrix theory. The nonlinear version materially improves performance in large universes.

These shrinkage estimators sharpen the inputs, but they do not change the underlying rule: mean-variance weights still scale as $w \propto \Sigma^{-1}\mu$, and the familiar pathologies can re-emerge when correlations shift in stressed markets. This motivates the second line of work, which drops optimisation altogether.

2.4 Hierarchical methods

López de Prado [2016] introduced *Hierarchical Risk Parity* (HRP), which constructs portfolio weights without ever inverting a covariance matrix. HRP proceeds in three steps. First, the correlation matrix is turned into a distance matrix and agglomerative

hierarchical clustering groups related assets. Second, the covariance matrix is “quasi-diagonalised” by permuting its rows and columns along the resulting tree, so that similar assets sit next to each other. Third, weights are produced by recursive bisection: the ordered list is repeatedly split into two contiguous sub-clusters, and at each split the less volatile sub-cluster receives the larger share, with inverse-variance weights used at the leaves. Because no matrix is inverted, HRP sidesteps the ill-conditioning that breaks mean-variance in large universes, and the resulting weights are approximately stable under small input perturbations. López de Prado [2018] develops the method at length and reports simulation evidence that HRP outperforms both minimum-variance and $1/N$ allocations when inputs are noisy.

HRP has two attractive properties: it is robust to estimation error, and it respects the tree structure of market co-movement. But robustness comes at a price. In calm markets, off-diagonal correlation structure is informative and stable, and a mean-variance optimiser can exploit it productively where HRP cannot. Ciciretti and Bucci [2022] documented this asymmetry on a large NYSE universe: a minimum-variance portfolio dominates HRP in calm periods, while HRP dominates during stressed ones. That empirical asymmetry is the central observation behind this thesis.

2.5 Regime detection

The assumption of stationary returns that underpins MPT is contradicted by a substantial empirical literature. Markets exhibit *regimes*: long stretches of low volatility and rising prices, interrupted by shorter and sharper episodes of high volatility and coincident drawdowns. Hamilton [1989] brought Markov-switching models into mainstream econometrics by modelling US GNP growth as being driven by an unobserved two-state Markov chain with state-dependent means. Hamilton’s model is a Hidden Markov Model (HMM) with an autoregressive emission, and the HMM has since become standard equipment for regime analysis in financial time series.

Applied to portfolio choice, regime models produced a memorable empirical finding. Ang and Bekaert [2002, 2004] documented a “bear-market correlation” effect: cross-asset correlations rise conditional on market stress, which erodes the diversification an investor would expect from unconditional correlation estimates. Guidolin and Timmermann [2007] extended the analysis to a multivariate four-regime model for stocks, bonds, and T-bills, and showed that the optimal allocation swings substantially across regimes.

A methodological point running through the applied literature concerns how the latent state is inferred. Rabiner [1989] provides the canonical reference on HMM estimation via the Baum-Welch algorithm and on state inference via the forward and Viterbi algorithms. As emphasised by Nystrup et al. [2018, 2020] and Bulla et al. [2011], the distinction between *filtered* state inference — which uses only information up to time t — and *smoothed* state inference — which uses the full sample — is consequential. Using smoothed states in a backtest is a subtle but severe form of look-ahead bias that can inflate apparent performance beyond recognition. This thesis uses filtered inference throughout; Section 3.4 returns to the point.

Simpler regime indicators are also in common use. Rules based on the VIX level or rolling realised volatility are standard in practice and will serve here as transparent baselines

against which the HMM must prove itself. These rules trade the HMM’s probabilistic state uncertainty and persistence structure for interpretability and near-zero computational cost.

2.6 Regime-switching portfolio construction

The natural next step is to let portfolio construction itself depend on the inferred regime. [Zhou and Yin \[2003\]](#) give an early analytical treatment, deriving the optimal mean-variance policy under a regime-switching diffusion. [Costa and Kwon \[2019\]](#) formulate a regime-dependent mean-variance problem and document out-of-sample gains over unconditional alternatives. [Nystrup et al. \[2018, 2020\]](#) provide extensive empirical evidence that *adapting* a single allocator to regime-conditional inputs beats static allocation across several asset classes.

Recent work has begun to exploit regime information more indirectly. [Wang et al. \[2020\]](#) use an HMM to rotate between factor models, switching the set of explanatory factors according to the inferred state. [Millea and Edalat \[2023\]](#) apply deep reinforcement learning to pick among several HRP variants on the fly. Both approaches share the spirit of the present thesis — adapting the choice of method to market conditions — but neither pairs a robust Markowitz method with a hierarchical method, and neither uses the HMM as a *discrete selector* rather than a continuous adapter.

2.7 Synthesis and research gap

Four claims emerge from the preceding sections and motivate the design in Chapter 3. First, mean-variance optimisation is fragile under estimation error, but a good covariance estimator — Ledoit-Wolf shrinkage being the natural choice — substantially narrows the gap to robust alternatives. Second, HRP is robust but can underperform in calm markets, where the off-diagonal correlation information it discards is reliable and productive. Third, market regimes are real and can be detected by an HMM, provided state inference is causal. Fourth, earlier work has combined regime detection with portfolio construction in several ways — regime-adaptive inputs, factor rotation, reinforcement learning over allocator variants — but, to the best of our knowledge, no paper has yet used an HMM as a *discrete selector* between a Ledoit-Wolf-shrunk mean-variance allocator and Hierarchical Risk Parity.

This thesis fills that gap. The working hypothesis is that the asymmetry documented by [Ciciretti and Bucci \[2022\]](#) is mediated by the market regime, and that an HMM-based selector can turn this observation into a deployable switching strategy. We do not claim that HMM switching must dominate its constituent strategies. A carefully documented *negative* result — that the switcher does not outperform either baseline after costs — would be informative in its own right, and the empirical design in Chapter 3 is built so that either conclusion is defensible.

Chapter 3

Methodology

The methodology has six pieces. Section 3.1 fixes notation and sets up the portfolio construction problem. Sections 3.2 and 3.3 describe the two candidate allocators: Ledoit-Wolf mean-variance and Hierarchical Risk Parity. Section 3.4 develops the Hidden Markov Model used for regime detection, paying particular attention to the filtering-versus-smoothing point raised in the literature review. Section 3.5 assembles these components into the switching framework, and Section 3.6 specifies the benchmarks against which it is evaluated.

3.1 Notation and portfolio problem setup

Let N be the number of risky assets in the investable universe, and let $r_{i,t}$ denote the log return on asset i in period t , stacked into $r_t = (r_{1,t}, \dots, r_{N,t})^\top$. A portfolio is a vector of weights $w_t = (w_{1,t}, \dots, w_{N,t})^\top$ chosen at the end of period t and held through period $t + 1$.

Three constraints are imposed throughout: long-only ($w_{i,t} \geq 0$), full investment ($\mathbf{1}^\top w_t = 1$), and a maximum-weight cap $w_{i,t} \leq w_{\max}$ set to $w_{\max} = 0.20$ in the primary specification. Long-only allocation is standard in practical portfolio applications; the weight cap reduces concentration and improves numerical stability.

The gross portfolio return over $[t, t + 1]$ is $R_{t+1}^p = w_t^\top r_{t+1}$. Turnover at each rebalance is

$$\tau_t = \frac{1}{2} \sum_{i=1}^N |w_{i,t} - w_{i,t-1}^{\text{drift}}|, \quad (3.1)$$

where $w_{i,t-1}^{\text{drift}}$ is the asset's weight immediately before rebalancing, after it has drifted from its previous target because of differential asset returns. Proportional transaction costs are charged at rate c per unit turnover, so the net portfolio return is $R_{t+1}^{p,\text{net}} = w_t^\top r_{t+1} - c \tau_t$. The primary specification uses $c = 10$ basis points; sensitivity to $c \in \{0, 5, 25\}$ bps is reported in the robustness section.

Rebalancing occurs monthly. All estimators are fitted on a rolling window of $W = 252$ trading days ending at the current rebalance date, so no information from beyond period t enters the portfolio held over $[t, t + 1]$. This point-in-time constraint is enforced in code and is the single most important feature of the backtesting infrastructure.

3.2 Mean-Variance with Ledoit-Wolf shrinkage

3.2.1 The sample covariance problem

Given a window of W centred return observations, the sample covariance is

$$S_t = \frac{1}{W-1} \sum_{s=t-W+1}^t (r_s - \bar{r})(r_s - \bar{r})^\top. \quad (3.2)$$

When N is of the same order as W — routine in large-universe applications — S_t is badly conditioned: its smallest eigenvalues are biased toward zero while its largest are biased upward (the Marchenko-Pastur phenomenon). Because unconstrained mean-variance weights scale as $w \propto S_t^{-1}\mu$, errors in the smallest eigenvalues are amplified in the inverse, and the optimiser loads heavily on the corresponding eigenvectors — generically directions of noise rather than signal. This is the mechanical origin of the instability diagnosed by [Michaud \[1989\]](#).

3.2.2 Shrinkage estimator and optimal intensity

[Ledoit and Wolf \[2004\]](#) address the problem with a linear shrinkage estimator that convexly combines the sample covariance with a structured target:

$$\hat{\Sigma}_{\text{LW},t} = \alpha_t^* F_t + (1 - \alpha_t^*) S_t. \quad (3.3)$$

The target F_t is the *constant-correlation* covariance: every off-diagonal correlation in S_t is replaced with the average off-diagonal correlation, and the covariance is rebuilt from the sample variances. F_t has $N + 1$ effective parameters — one correlation and N variances — and is immune to the small-eigenvalue pathology at the cost of bias.

The shrinkage intensity $\alpha_t^* \in [0, 1]$ is chosen to minimise the expected Frobenius loss $\mathbb{E}[\|\hat{\Sigma}_{\text{LW},t} - \Sigma_t\|_F^2]$, which yields a closed-form expression in terms of sample moments; the full derivation is given in [Ledoit and Wolf \[2004\]](#). Intuitively, α_t^* is large when S_t is noisy (short windows or large universes) and small when S_t is stable. The estimator is therefore self-regularising: it shrinks hardest when shrinking matters most.

3.2.3 Optimisation problem

Given $\hat{\Sigma}_{\text{LW},t}$, we solve the minimum-variance problem

$$w_t^{\text{MV}} = \arg \min_w w^\top \hat{\Sigma}_{\text{LW},t} w \quad \text{s.t.} \quad \mathbf{1}^\top w = 1, \quad 0 \leq w \leq w_{\max} \mathbf{1}. \quad (3.4)$$

We use the minimum-variance rather than the maximum-Sharpe formulation for two reasons. First, expected returns are notoriously hard to estimate at monthly horizons on a five-year window [[Chopra and Ziemba, 1993](#)], and passing a noisy $\hat{\mu}$ into the objective re-introduces the instability the shrinkage was designed to remove. Second, dropping $\hat{\mu}$ puts MV and HRP on equal footing: both allocators are then driven only by the second moment, which makes the head-to-head comparison unambiguous. Problem (3.4) is a convex quadratic program and is solved with `cvxpy` under its default solver. For the shrinkage ablation, the same program is solved with S_t in place of $\hat{\Sigma}_{\text{LW},t}$.

3.3 Hierarchical Risk Parity

HRP produces weights without inverting a covariance matrix. We follow [López de Prado \[2016\]](#).

3.3.1 Distance metric and hierarchical clustering

Let ρ_t be the $N \times N$ sample correlation matrix on the estimation window. Define the pairwise distance

$$d_{ij,t} = \sqrt{\frac{1}{2}(1 - \rho_{ij,t})}, \quad (3.5)$$

which is a proper metric on $[0, 1]$: $d_{ij,t} = 0$ when $\rho_{ij,t} = 1$, and $d_{ij,t} = 1$ when $\rho_{ij,t} = -1$. Agglomerative hierarchical clustering with single linkage is then applied to d_t , producing a dendrogram — a binary tree of nested asset groupings. Single linkage is the standard choice in the HRP literature [[López de Prado, 2016](#)]; sensitivity to the linkage rule is examined in the robustness section.

3.3.2 Quasi-diagonalisation

The dendrogram induces a permutation of the N assets that places similar assets next to each other. Applying this permutation to the rows and columns of the sample covariance $\hat{\Sigma}_t$ yields a *quasi-diagonal* matrix with large values clustered near the main diagonal. This step does not change the information content of the covariance; it only reorders the assets so that the subsequent recursive split operates on coherent groups.

3.3.3 Recursive bisection

Starting from the full reordered list of N assets, the algorithm recursively bisects the list into two contiguous sub-clusters L and R . For each sub-cluster C , the variance of the inverse-variance-weighted portfolio of its constituents is

$$V_C = \tilde{w}_C^\top \hat{\Sigma}_{C,t} \tilde{w}_C, \quad \tilde{w}_{i,C} \propto \hat{\sigma}_{i,t}^{-2}. \quad (3.6)$$

A single split factor is then applied. The weight assigned to the left sub-cluster is

$$\alpha_L = 1 - \frac{V_L}{V_L + V_R}, \quad (3.7)$$

and the weight assigned to R is $1 - \alpha_L$; the less volatile sub-cluster receives the larger share. The procedure recurses within each sub-cluster until singleton leaves are reached. Each asset-level weight $w_{i,t}^{\text{HRP}}$ is the product of the split factors along the path from the root of the tree to that leaf.

Two properties of HRP matter for this thesis. First, the algorithm never inverts a covariance matrix, so the numerical issues caused by near-singular $\hat{\Sigma}_t$ do not arise. Second, risk is balanced hierarchically — within and across clusters — rather than through explicit optimisation. The cost is that HRP uses off-diagonal correlation information only indirectly, through the clustering step. When the off-diagonals are reliable, as they often are in calm markets, a mean-variance optimiser can do better [[Ciciretti and Bucci, 2022](#)].

3.4 Hidden Markov Model for regime detection

3.4.1 Model specification

The market regime is modelled as a discrete, two-state, first-order Markov chain $\{s_t\}_{t \geq 0}$, $s_t \in \{0, 1\}$, with transition matrix

$$P = \begin{pmatrix} p_{00} & p_{01} \\ p_{10} & p_{11} \end{pmatrix}, \quad p_{ij} = \Pr(s_{t+1} = j \mid s_t = i). \quad (3.8)$$

Conditional on $s_t = i$, the observed feature vector x_t is drawn from a multivariate Gaussian $\mathcal{N}(\mu_i, \Sigma_i)$. The parameter set is $\theta = (P, \mu_0, \mu_1, \Sigma_0, \Sigma_1, \pi_0)$, where π_0 is the initial state distribution.

Two states reflect the common empirical finding that equity markets alternate mainly between a “calm” regime and a “turbulent” one; three-state specifications are considered in robustness checks. The HMM likelihood is invariant to permutations of state labels, so we fix labels ex post by sorting states on their mean VIX level: state 0 is the low-volatility regime and state 1 the high-volatility one.

3.4.2 Feature selection

Following the pre-registered design and Nystrup et al. [2018], the feature vector x_t is two-dimensional: the level of the CBOE VIX index and its log change. These features carry information about the current volatility environment, have economically interpretable scales, and are observable at the end of each trading day, which allows real-time state inference. Keeping the feature set small — two variables on a long sample — also limits overfitting and keeps the estimated regime dynamics easy to read.

3.4.3 Estimation via Expectation-Maximisation

Parameters are estimated by maximum likelihood via the Baum-Welch algorithm, which is Expectation-Maximisation applied to hidden-state models [Rabiner, 1989]. Because the HMM likelihood is multimodal, EM is run from ten random initialisations, and the fit with the highest converged log-likelihood is kept. We use the `hmmlearn` implementation.

3.4.4 Online state inference: filtering versus smoothing

The state sequence used for portfolio decisions must be inferred *causally*: using only information available at or before time t . This is the *filtering* distribution $\Pr(s_t \mid x_{1:t}, \theta)$, computed by the forward algorithm. The alternative *smoothing* distribution $\Pr(s_t \mid x_{1:T}, \theta)$ conditions on the full sample and is, by construction, unavailable in real time. Using smoothed states in a backtest is a subtle but severe form of look-ahead bias that can inflate measured performance beyond recognition [Nystrup et al., 2018]. We use filtered probabilities exclusively for out-of-sample state assignment; filtered and smoothed sequences are compared post hoc as a diagnostic only.

3.4.5 Rolling re-estimation

HMM parameters are re-estimated on an expanding window every twelve months, using all data available up to the re-estimation date. This reflects the possibility that regime

dynamics themselves evolve slowly over time, while avoiding the excess variance of more frequent refitting. Between re-estimations, the filtering recursion is updated each trading day using the most recently fitted parameters.

3.5 The switching framework

3.5.1 Regime classification rule

At each month-end rebalance date t , we compute the filtered state probabilities $\Pr(s_t = i \mid x_{1:t}, \hat{\theta}_t)$ from the most recently fitted HMM. The regime is classified deterministically as

$$\hat{s}_t = \arg \max_{i \in \{0,1\}} \Pr(s_t = i \mid x_{1:t}, \hat{\theta}_t). \quad (3.9)$$

A probabilistic mixing variant, in which the portfolio is a convex combination of w_t^{MV} and w_t^{HRP} weighted by the filtered state probabilities, is considered separately in the robustness section.

3.5.2 Allocator assignment per regime

The switching rule is

$$w_t^{\text{switch}} = \begin{cases} w_t^{\text{MV}} & \text{if } \hat{s}_t = 0 \text{ (calm regime),} \\ w_t^{\text{HRP}} & \text{if } \hat{s}_t = 1 \text{ (turbulent regime).} \end{cases} \quad (3.10)$$

This mapping encodes the thesis’s central hypothesis, following the asymmetry reported by [Ciciretti and Bucci \[2022\]](#). When the market is calm, off-diagonal correlation information is reliable and productive, so LW-MV is the better bet. When the market is turbulent, correlations are less stable — contaminated by the bear-market correlation effect of [Ang and Bekaert \[2002\]](#) — and the robustness of HRP is worth its lower expected Sharpe in calm states.

3.5.3 Rebalancing logic and cost handling

At each rebalance date, both allocators are constructed and the regime is inferred. The switching rule picks one set of weights as the target, turnover is computed against the actual drifted weights from the previous rebalance, and transaction costs are applied at the rate set in [Section 3.1](#).

A well-known failure mode of switching strategies is *whipsawing*: frequent regime flips driven by transient feature noise, each one triggering a costly rebalance. The HMM’s explicit persistence modelling through the transition matrix P provides a natural brake: the diagonal probabilities p_{ii} are typically estimated well above 0.9, which makes the filtered distribution sticky and insensitive to day-to-day noise. The actual turnover cost of switching is nonetheless measured in the results chapter and reported as a separate line of evidence.

3.6 Benchmark strategies

The HMM-switching strategy is compared against the following benchmarks, each constructed on the same data, estimation windows, and transaction-cost model. All benchmarks are subject to the same long-only, full-investment, and weight-cap constraints as the main strategy.

- **1/ N equal-weight** [DeMiguel et al., 2009]: $w_{i,t} = 1/N$ for all i and t . No estimated parameters, and the most parsimonious baseline.
- **Standalone Ledoit-Wolf MV**: w_t^{MV} applied every period, irrespective of regime.
- **Standalone HRP**: w_t^{HRP} applied every period, irrespective of regime.
- **Sample-covariance MV**: the minimum-variance program solved with S_t in place of $\hat{\Sigma}_{\text{LW},t}$. Included for the shrinkage ablation, isolating the contribution of Ledoit-Wolf shrinkage.
- **VIX-threshold selector**: the same switching rule as in Section 3.5, except that the regime is classified by whether the current VIX level is above its median over the estimation window. This baseline isolates what the HMM adds over a transparent, near-costless alternative.

Performance is evaluated on annualised return and volatility, Sharpe and Sortino ratios, maximum drawdown, Calmar ratio, and turnover. Statistical significance of Sharpe differences between the switching strategy and each benchmark is assessed with the Ledoit and Wolf [2008] robust Sharpe test, and confidence intervals are built using the stationary bootstrap of Politis and Romano [1994]. Where space permits, the Deflated Sharpe Ratio of Bailey and López de Prado [2014] is reported to adjust for the multiplicity of strategies evaluated.

Chapter 4

Results

Present your findings, possibly with figures and tables. For example:

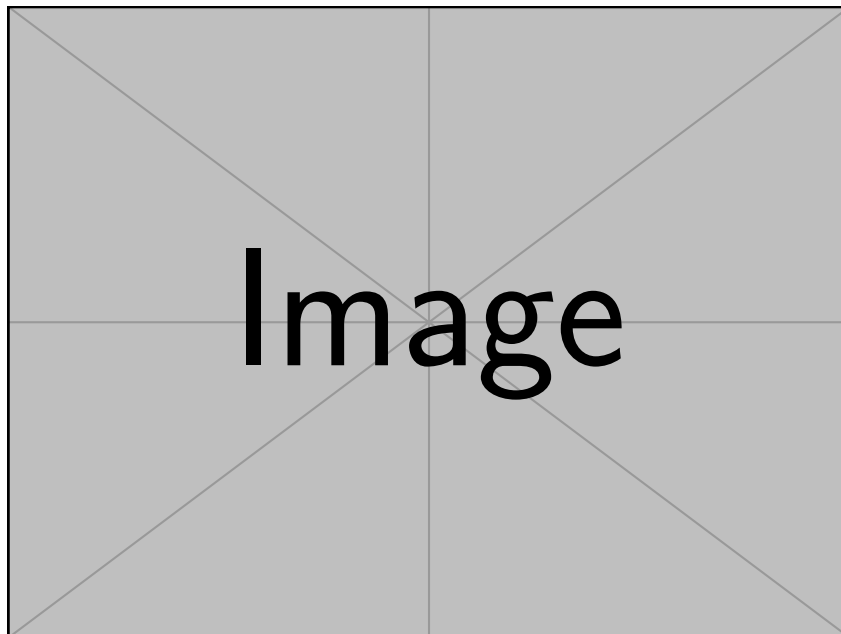


Figure 4.1: Example figure caption.

Chapter 5

Discussion

Interpret results, compare with literature, discuss limitations.

Chapter 6

Conclusion

Summarize contributions, implications, and future work.

Appendix A

Additional Data

Include raw data, code snippets, or extended tables here.

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